

Chapter 21

Waves in materials

21.1 Wave equation in a medium

Consider a material with permeability μ , permittivity ϵ , and conductivity σ , with no external sources. The Maxwell equations read

$$\vec{\nabla} \times \vec{H} = \frac{\epsilon}{c} \frac{\partial \vec{E}}{\partial t} + \frac{4\pi\sigma}{c} \vec{E} \quad , \quad \vec{\nabla} \times \vec{E} = -\frac{\mu}{c} \frac{\partial \vec{H}}{\partial t} \quad (21.1)$$

along with

$$\vec{\nabla} \cdot \vec{H} = 0 \quad , \quad \vec{\nabla} \cdot \vec{E} = 0 \quad (21.2)$$

Combining the two curl equations leads to

$$\vec{\nabla}^2 \vec{H} = \frac{\mu\epsilon}{c^2} \frac{\partial^2 \vec{H}}{\partial t^2} + \frac{4\pi\mu\sigma}{c^2} \frac{\partial \vec{H}}{\partial t} \quad (21.3)$$

(and the same equation for $\vec{E}$) – a wave equation. The associated wave speed and index of refraction are

$$c' \equiv \frac{c}{\sqrt{\mu\epsilon}} \quad , \quad n \equiv \frac{c}{c'} = \sqrt{\mu\epsilon} \quad (21.4)$$

(for $\mu \sim 1$, $n \sim \sqrt{\epsilon}$).

21.2 Elementary waves and the dispersion relation

Substitute elementary waves

$$\vec{E} = \vec{E}_0 e^{i(\vec{K} \cdot \vec{x} - \omega t)} \quad , \quad \vec{H} = \vec{H}_0 e^{i(\vec{K} \cdot \vec{x} - \omega t)} \quad (21.5)$$

The first curl equation gives

$$\begin{aligned} i \vec{K} \times \vec{H}_0 &= -\frac{\epsilon}{c} i \omega \vec{E}_0 + \frac{4\pi\sigma}{c} \vec{E}_0 \\ &= -i \frac{\omega}{c} \eta \vec{E}_0 \end{aligned} \quad (21.6)$$

where

$$\eta \equiv \epsilon + \frac{4\pi i \sigma}{\omega} \quad (21.7)$$

is a complex permittivity for $\sigma \neq 0$. Similarly,

$$\vec{K} \times \vec{E}_0 = \frac{\omega}{c} \mu \vec{H}_0 \quad , \quad \vec{K} \cdot \vec{E}_0 = \vec{K} \cdot \vec{H}_0 = 0 \quad (21.8)$$

(note that $|\vec{E}| \neq |\vec{H}|$ in a medium). Crossing once more with $\vec{K}$,

$$-\vec{K}^2 \vec{H}_0 = -\frac{\omega}{c} \eta \vec{K} \times \vec{E}_0 \Rightarrow \vec{K}^2 = \frac{\omega^2}{c^2} \eta \mu \equiv K^2 \quad (21.9)$$

a new dispersion relation. In general, K is complex

$$K = \alpha + i\beta = \sqrt{\frac{\mu\epsilon}{2}} \frac{\omega}{c} \left[\sqrt{\Delta + 1} + i \sqrt{\Delta - 1} \right] \quad (21.10)$$

where

$$\Delta \equiv \sqrt{1 + \left(\frac{4\pi\sigma}{\omega\epsilon} \right)^2} \quad [?] \quad (21.11)$$

so that

$$|K| = \sqrt{\epsilon\mu} \frac{\omega}{c} \Delta^{1/2} [?] \quad (21.12)$$

and the phase of K is

$$\tan^{-1} \frac{\beta}{\alpha} = \frac{1}{2} \tan^{-1} \frac{4\pi\sigma}{\omega\epsilon} \quad (21.13)$$

The wave then goes as

$$e^{i(Kx-\omega t)} = e^{i(\alpha x-\omega t)} e^{-\beta x} \quad (21.14)$$

where the last factor is a decay ($\beta > 0$): the penetration depth is $\sim 1/\beta$, while the wavelength is $\lambda = 2\pi/\alpha$. Over a wavelength, the fields decay as

$$e^{-2\pi\beta/\alpha} \quad (21.15)$$

Generally, the conductivity is frequency dependent

$$\sigma \rightarrow \sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau} \quad (21.16)$$

(the Drude model), and $\epsilon \rightarrow \epsilon(\omega)$ as well – so all these effects are frequency dependent.

21.3 Fresnel's law

Consider an interface at $z = 0$ between a medium (ϵ, μ) at $z < 0$ and a medium (ϵ', μ') at $z > 0$, with $\sigma = 0$, as depicted in the Figure: an incident wave $\vec{K}$ at angle α from the normal $\hat{n}$, a reflected wave $\vec{K}''$ at angle γ , and a transmitted wave $\vec{K}'$ at angle β . The three waves are

$$\vec{E} = \vec{E}_0 e^{i\vec{K}\cdot\vec{x}} \quad , \quad \vec{B} = \sqrt{\mu\epsilon} \frac{\vec{K} \times \vec{E}}{K} \quad (21.17)$$

$$\vec{E}' = \vec{E}'_0 e^{i\vec{K}'\cdot\vec{x}} \quad , \quad \vec{B}' = \sqrt{\mu'\epsilon'} \frac{\vec{K}' \times \vec{E}'}{K'} \quad (21.18)$$

$$\vec{E}'' = \vec{E}''_0 e^{i\vec{K}''\cdot\vec{x}} \quad , \quad \vec{B}'' = \sqrt{\mu\epsilon} \frac{\vec{K}'' \times \vec{E}''}{K''} \quad (21.19)$$

We have continuity conditions on

$$(1) E_t \quad , \quad (2) B_n \quad , \quad (3) H_t \quad , \quad (4) D_n \quad (21.20)$$

holding for all t and all x, y at $z = 0$, which requires

$$K_x = K'_x = K''_x \quad (21.21)$$

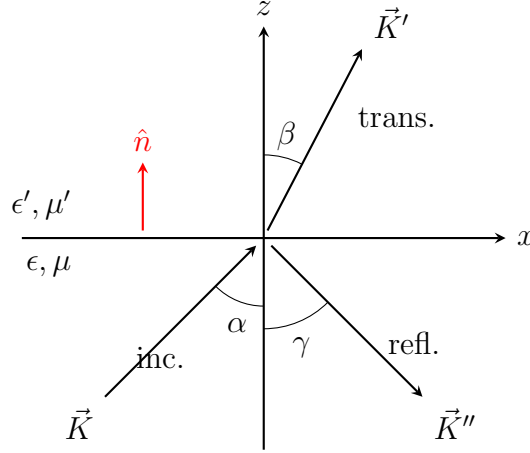


Figure 21.1: Incident, reflected, and transmitted waves at an interface at $z = 0$ between two media, with normal $\hat{n}$.

This implies

$$\omega = \omega' = \omega'' \Rightarrow |\vec{K}| = |\vec{K}''| = \frac{\omega}{c} \sqrt{\epsilon \mu} \quad (21.22)$$

and

$$\vec{K} \cdot \vec{r} = \vec{K}' \cdot \vec{r} = \vec{K}'' \cdot \vec{r} \Big|_{z=0} \quad (21.23)$$

that is

$$|\vec{K}| \sin \alpha = |\vec{K}'| \sin \beta = |\vec{K}''| \sin \gamma \quad (21.24)$$

leading to

$$\alpha = \gamma \quad \text{and} \quad \frac{\sin \alpha}{\sin \beta} = \frac{|\vec{K}'|}{|\vec{K}|} = \frac{\sqrt{\mu' \epsilon'}}{\sqrt{\mu \epsilon}} = \frac{n'}{n} \quad (21.25)$$

The four continuity conditions then read

$$\begin{aligned} (\vec{E}_0 + \vec{E}_0'' - \vec{E}_0') \times \hat{n} &= 0 \\ (\vec{K} \times \vec{E}_0 + \vec{K}'' \times \vec{E}_0'' - \vec{K}' \times \vec{E}_0') \cdot \hat{n} &= 0 \\ \left[\frac{1}{\mu} (\vec{K} \times \vec{E}_0 + \vec{K}'' \times \vec{E}_0'') - \frac{1}{\mu'} (\vec{K}' \times \vec{E}_0') \right] \times \hat{n} &= 0 \\ \left(\epsilon (\vec{E}_0 + \vec{E}_0'') - \epsilon' \vec{E}_0' \right) \cdot \hat{n} &= 0 \end{aligned} \quad (21.26)$$

21.4 The two polarizations

Two cases arise. First, take $\vec{E}_0$ perpendicular to the plane of incidence. The conditions become

$$E_0 + E_0'' - E_0' = 0 \quad , \quad (E_0 - E_0'') \sqrt{\frac{\epsilon}{\mu}} \cos \alpha - E_0' \sqrt{\frac{\epsilon'}{\mu'}} \cos \beta = 0 \quad (21.27)$$

leading to

$$\frac{E_0''}{E_0} = \frac{1 - \frac{\mu}{\mu'} \frac{\tan \alpha}{\tan \beta}}{1 + \frac{\mu}{\mu'} \frac{\tan \alpha}{\tan \beta}} \simeq \frac{\sin(\beta - \alpha)}{\sin(\alpha + \beta)} \quad (21.28)$$

and

$$\frac{E_0'}{E_0} = \frac{2}{1 + \frac{\mu}{\mu'} \frac{\tan \alpha}{\tan \beta}} \simeq \frac{2 \sin \beta \cos \alpha}{\sin(\alpha + \beta)} \quad (21.29)$$

where the $\simeq$ forms hold for $\mu \sim \mu'$.

Second, take $\vec{E}_0$ in the plane of incidence. Then

$$\frac{E_0''}{E_0} = \frac{1 - \frac{\epsilon}{\epsilon'} \frac{\tan \alpha}{\tan \beta}}{1 + \frac{\epsilon}{\epsilon'} \frac{\tan \alpha}{\tan \beta}} \simeq \frac{\tan(\alpha - \beta)}{\tan(\alpha + \beta)} \quad (21.30)$$

and

$$\frac{E_0'}{E_0} = \frac{2 \frac{n'}{n} \frac{\epsilon}{\epsilon'}}{1 + \frac{\epsilon}{\epsilon'} \frac{\tan \alpha}{\tan \beta}} \simeq \frac{2 \sin \beta \cos \alpha}{\sin(\alpha + \beta) \cos(\alpha - \beta)} \quad (21.31)$$

again with the $\simeq$ forms for $\mu \sim \mu'$.

21.5 Total internal reflection

Note the following special cases:

- $n = n'$ implies $E_0'' = 0$: no reflection.
- $n' \rightarrow \infty$ implies $E_0' = 0$: no transmission.
- $\beta_0 = \pi/2$ at the angle $\sin \alpha_0 = n'/n$ (requiring $n' < n$): total reflection.

In the last case, for $\alpha > \alpha_0$,

$$\begin{aligned}\cos \beta &= \sqrt{1 - \left(\frac{\sin \alpha}{\sin \alpha_0} \right)^2} \\ &= i \sqrt{\left(\frac{\sin \alpha}{\sin \alpha_0} \right)^2 - 1}\end{aligned}\tag{21.32}$$

so that the transmitted wave becomes

$$e^{i(|\vec{K}'| x \sin \beta + |\vec{K}'| z \cos \beta)} = e^{i|\vec{K}'| x \frac{n}{n'} \sin \alpha} e^{-|\vec{K}'| z \sqrt{\left(\frac{\sin \alpha}{\sin \alpha_0} \right)^2 - 1}}\tag{21.33}$$

a standing wave along the interface times a decaying factor: there is no refraction, with penetration depth $\sim \lambda$ and $\vec{S} \cdot \hat{n} = 0$.